

Sports Injury Risk Detection — Complete Zero-Knowledge Presentation & Viva Master Guide

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System Overview: The Sports Injury Risk Detection and Prevention System (Motion IQ) is an AI-powered biomechanical movement analysis, injury risk detection, and rehabilitation tracking platform. It pairs a **React 18 (Vite) Single Page Application** hosted on **Vercel** with a **FastAPI (Python 3.14)** backend hosted on **Render**, powered by **Google MediaPipe BlazePose (33 3D Keypoints)**, **OpenCV**, **XGBoost / Random Forest ML models**, **SQLAlchemy ORM**, and **ReportLab PDF Engine**.

SECTION 1: PROJECT IN ONE MINUTE

What Problem Are We Solving?

In sports science and physiotherapy, identifying faulty movement patterns (such as knee valgus collapse or leg asymmetry) before an injury occurs is critical. Traditional biomechanics laboratories require expensive multi-camera setups, body-attached sensors, and force plates costing tens of thousands of dollars.

Why Is This Problem Important?

ACL tears, hamstring strains, and joint injuries cost athletes entire seasons and require months of rehabilitation. Early detection of biomechanical deficits allows coaches to prescribe corrective exercises before severe injuries occur.

What Is Our Solution?

Our application enables a coach or athlete to record a video on any normal smartphone, upload it to a web application, and receive instant 3D joint telemetry, limb symmetry scores, AI-driven injury risk predictions, and a downloadable clinical PDF report—with **zero hardware or sensors required**.

Who Uses The System?

- * **Coaches & Trainers:** To monitor squad movement quality and injury risk.
- * **Physiotherapists:** To track rehabilitation progress and joint range of motion over time.
- * **Athletes:** To view personal movement feedback and corrective exercise prescriptions.

Input, Processing & Output

- * **User Input:** Smartphone exercise video file (.mp4, .mov, .webm), athlete profile details (age, sport, position), and exercise activity type (squatting, jumping, running).
- * **System Processing:** OpenCV decodes video frames, MediaPipe BlazePose detects 33 3D body keypoints, 3D vector geometry computes joint angles, and Random Forest / XGBoost ML models predict risk probability.
- * **System Output:** Interactive movement dashboard, 0–100% Injury Risk Score, risk tier classification (LOW, MODERATE, HIGH, CRITICAL), bilateral symmetry scores, and a downloadable multi-page clinical PDF report.

SECTION 2: THE COMPLETE SYSTEM IN ONE PICTURE

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USER BROWSER (React SPA on Vercel)



■ HTTPS REST API Requests (Authorization: Bearer <JWT>)

RENDER CLOUD BACKEND (FastAPI / Uvicorn ASGI Server)

[illegible]

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OPENCV & MEDIAPIPE ENGINE MACHINE LEARNING ENGINE

- Video frame decoding - Random Forest & XGBoost
- 640px Downsampling - 23 Kinematic Features
- 33 3D Pose Landmarks - Risk Score (.predict_proba)
- Vector Dot-Product Math - Rehabilitation Classifier

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SOLALCHEMY ORM DATABASE

(SOLite locally / PostgreSQL in Prod)

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REPORTLAB PDF GENERATOR

(Binary Clinical PDF Response Stream)

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SECTION 3: MOST IMPORTANT FLOW #1 — "WHAT HAPPENS WHEN I PRESS A BUTTON?"

Visual Execution Chain

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USER CLICKS BUTTON ■■> REACT EVENT HANDLER ■■> FRONTEND FUNCTION ■■> API REQUEST ■■> NETWORK ■■> FASTAPI ROUTE ■■> PYSCHEMA VALIDATION ■■> BACKEND SERVICE ■■> DB/CALCULATION ■■> API RESPONSE ■■> REACT STATE UPDATE ■■> VIRTUAL DOM RE-RENDER ■■> USER SEES RESULT

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Detailed 10-Step Breakdown in Simple English:

1. **User Clicks Button:** User clicks a UI button (e.g. "Add Athlete"). Browser captures the DOM click event.
2. **React Event Handler:** React calls the function bound to `onClick` in `frontend/src/main.jsx`.
3. **Frontend Function Runs:** Function reads user inputs stored in React `useState` variables.
4. **API Request Created:** `api(path, options)` constructs HTTP request, attaching `Authorization: Bearer <token>` from `localStorage`.
5. **Network Travel:** Browser sends HTTP POST/GET request over network to FastAPI backend URL.
6. **FastAPI Route Interception:** FastAPI matches route in `backend/app/routes/` and injects `get_current_user` dependency.
7. **Pydantic Validation:** Backend validates payload against Pydantic schema (`AthleteCreate`).

8. **Backend Logic & Database:** Python service executes database write via SQLAlchemy (`db.add()`, `db.commit()`).
9. **Backend Response:** FastAPI serializes database model to JSON and returns HTTP 200 OK.
10. **React State & Screen Update:** React `api()` promise resolves, calling `setAthletes()`. React re-renders component on screen.

SECTION 4: MOST IMPORTANT FLOW #2 — "WHAT HAPPENS TO THE VIDEO?"

Visual Video Pipeline

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MP4 VIDEO ■■> UPLOAD ROUTE ■■> DISK STORAGE ■■> OPENCV CAPTURE ■■> 640px DOWNSAMPLING ■■> MEDIAPIPE BLAZEPOSE ■■> 33 3D LANDMARKS ■■> VECTOR DOT MATH ■■> JOINT ANGLE SERIES ■■> ROM & SYMMETRY ■■> ML INFERENCE ■■> SQLITE DB ■■> REPORTLAB PDF

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Story of Video Analysis:

1. **Upload:** User selects `.mp4` file. React sends file via `multipart/form-data` to `/api/videos/upload-and-analyze`.
2. **Storage:** Backend writes video bytes to server disk (`uploads/videos/{uuid}.mp4`).
3. **OpenCV Decoding:** OpenCV `cv2.VideoCapture` opens the video stream, extracting FPS, width, height, and frame count.
4. **Frame Sampling:** Downsamples images wider than 640px to 640px for 10x faster MediaPipe inference.
5. **MediaPipe Pose Extraction:** MediaPipe BlazePose extracts 33 3D skeletal landmarks $(x, y, z, \text{visibility})$ per frame.
6. **Vector Geometry Math:** Computes 3D interior joint angles using vector dot-product cosine inverse formula $\theta = \arccos\left(\frac{\vec{u} \cdot \vec{v}}{|\vec{u}| |\vec{v}|}\right) \times \frac{180}{\pi}$.
7. **Biomechanical Rollup:** Computes Range of Motion ($\max - \min$), Bilateral Limb Symmetry, Trunk Lean, and Consistency.
8. **ML Model Risk Score:** 23 kinematic features are passed into Random Forest `.predict_proba()` to output 0–100% Risk Score.
9. **DB & PDF Output:** Result is stored in database via SQLAlchemy. Clicking "Download Report" invokes ReportLab PDF engine.

SECTION 5: WHAT HAPPENS WHEN I OPEN THE WEBSITE?

Production Step-by-Step Sequence:

1. User enters website URL (`https://...vercel.app`).
2. Browser sends HTTPS GET request to Vercel CDN.
3. Vercel serves `index.html` and bundled JavaScript (`dist/assets/index.js`).
4. Browser parses JavaScript and boots React 18 Single Page Application.
5. React initializes `AuthContext`, checks `localStorage.getItem('sir_auth')`.
6. If token exists, React sends initial background HTTP GET request to `/api/auth/me` or `/api/athletes`.
7. Request travels over HTTPS to Render backend (`https://injury-prediction-backend.onrender.com`).
8. If Render backend is sleeping (free tier inactive), Render initiates a cold start (takes 10–25 seconds).
9. FastAPI receives request, validates session token against database `sessions` table.
10. Backend returns user profile JSON. React updates `currentUser` state and renders Coach Dashboard.

SECTION 6: DOES THE BACKEND START WHEN I OPEN THE WEBSITE?

- * **Does opening Vercel start FastAPI?**: No. Vercel only hosts the frontend static files.
- * **Is FastAPI always running?**: In production on Render, the backend process runs independently of the browser. On free hosting, Render puts inactive backend instances to sleep after 15 minutes of inactivity.
- * **What happens if backend is sleeping?**: The first API request triggers a Render cold start. The browser waits until Uvicorn boots. Our frontend fetch wrapper includes timeout handling and retry notices.
- * **Who starts Uvicorn?**: Render starts Uvicorn via `backend/Procfile` command: `web: uvicorn app.main:app --host 0.0.0.0 --port $PORT`.
- * **Does every user start a new backend?**: No. A single FastAPI process handles thousands of concurrent user requests asynchronously.

SECTION 7: BROWSER VS FRONTEND VS BACKEND VS DATABASE

- * **Browser**: The software running on the user's laptop or phone (Chrome, Safari). It renders HTML/CSS and executes JavaScript.
- * **Frontend (React SPA)**: The visual user interface code running inside the browser. Handles buttons, forms, and charts.
- * **Backend (FastAPI)**: The server application running Python code in the cloud. Executes computer vision, ML models, and security validation.
- * **Database (SQLAlchemy + SQLite/PostgreSQL)**: The persistent storage memory. Stores user accounts, athlete profiles, and video analysis history.
- * **API (Application Programming Interface)**: The structured HTTP bridge allowing Frontend and Backend to talk to each other.
- * **CORS (Cross-Origin Resource Sharing)**: Security rules allowing the Vercel frontend domain to make API requests to the Render backend domain.

SECTION 8: COMPLETE ACTION-BY-ACTION TRACE

ACTION 1: User Log In

- * **User Action**: Enters email and password, clicks "Login".
- * **UI Component**: `<AuthScreen>` in `frontend/src/main.jsx`.
- * **Event Handler**: `handleSubmit(e)`.
- * **Frontend Function**: Calls `api('/api/auth/login', { method: 'POST', body: JSON.stringify({ email, password }) })`.
- * **API Route**: `POST /api/auth/login` in `backend/app/routes/auth.py`.
- * **Backend Validation**: Pydantic `UserLogin` schema checks email format.
- * **Backend Operation**: Queries database `users` table, verifies password via Passlib Bcrypt salt hash.
- * **Database Write**: Inserts session row into `sessions` table.
- * **Response**: Returns `{ token: "...", user: { user_id: "...", name: "...", role: "coach" } }`.
- * **State Update**: Saves token in `localStorage`, calls `setAuthenticated(true)`, `setCurrentUser(user)`.
- * **Screen Output**: Redirects to Coach Dashboard and displays welcome toast notification.

ACTION 2: Add New Athlete

- * **User Action:** Clicks "+ Add Athlete", fills form (Name: "Alex Rivera", Age: 22, Sport: "Football"), clicks "Save Athlete".
- * **UI Component:** `<AthleteModal>` in `frontend/src/main.jsx`.
- * **Event Handler:** `handleSaveAthlete()`.
- * **API Route:** `POST /api/athletes` in `backend/app/routes/athletes.py`.
- * **Backend Operation:** `get_current_user` validates Bearer JWT token. Creates SQLAlchemy `Athlete(user_id=current_user.id, name="Alex Rivera", ...)`. Executes `db.add()` and `db.commit()`.
- * **Response:** Returns created `Athlete` schema JSON.
- * **State Update:** React executes `setAthletes(prev => [newAthlete, ...prev])`.
- * **Screen Output:** Roster table re-renders displaying "Alex Rivera" in the athlete matrix.

ACTION 3: Upload & Analyze Video

- * **User Action:** Selects `squat.mp4`, chooses activity "squatting", clicks "Analyze Movement".
- * **UI Component:** `<VideoAnalysisTab>` in `frontend/src/main.jsx`.
- * **Event Handler:** `handleUploadAndAnalyze()`.
- * **API Route:** `POST /api/videos/upload-and-analyze` in `backend/app/routes/videos.py`.
- * **Backend Operation:**
 1. Saves file to `uploads/videos/{uuid}.mp4`.
 2. `process_video()` opens OpenCV stream, downsamples to 640px, extracts MediaPipe 33 3D keypoints for 30 frames.
 3. `run_full_biomechanics()` calculates joint angles, ROM, knee symmetry, and trunk lean.
 4. `compute_risk()` runs Random Forest ML model `.predict_proba()` to compute 78% Risk Score (HIGH RISK).
 5. Writes `MovementAnalysis` and `RiskResult` rows to database.
- * **Response:** Returns full `AnalysisResult` JSON.
- * **State Update:** React updates `analysisState` with biomechanics JSON and risk score.
- * **Screen Output:** Displays 3D joint skeleton preview, risk gauge badge, and "Download Clinical PDF Report" button.

ACTION 4: Download Clinical PDF Report

- * **User Action:** Clicks "Download Clinical PDF Report".
- * **API Route:** `GET /api/reports/{analysis_id}` in `backend/app/routes/reports.py`.
- * **Backend Operation:** Queries analysis session, fetches athlete profile and risk results, invokes `ReportLab generate_pdf_report()` to build PDF on disk (`results/reports/{id}_report.pdf`).
- * **Response:** Returns `FileResponse` binary PDF stream (Content-Type: `application/pdf`).
- * **Screen Output:** Browser opens native PDF download dialog.

ACTION 5: Toggle UI Color Theme (8 Themes)

- * **User Action:** Selects theme from dropdown (e.g. "Dark Elite", "Slate Pro", "Neon Cyber").
- * **UI Component:** `<ThemeSelector>` in `frontend/src/main.jsx`.
- * **State Update:** React `ThemeContext` updates `theme` state and sets `document.documentElement.setAttribute('data-theme', theme)`.
- * **Screen Output:** CSS custom variables dynamically update colors across the DOM instantly without server calls.

SECTION 9: EVERY API EXPLAINED LIKE A BEGINNER

* **An API is like a waiter in a restaurant:** The customer (Frontend) looks at the menu, tells the waiter (API) their order, the waiter takes the order to the kitchen (Backend/Database), and brings back the prepared food (JSON Data).

HTTP Method	API Route	Purpose	Who Calls It	Request Body	Response Payload
POST	/api/auth/register	Create user account	Signup Screen	{ name, email, password, role }	{ token, user }
POST	/api/auth/login	Authenticate user	Login Screen	{ email, password }	{ token, user }
GET	/api/auth/me	Verify session token	App Startup	Headers: Bearer <token>	{ user_id, name, email, role }
GET	/api/athletes	List coach's athletes	Dashboard	Headers: Bearer <token>	[{ athlete_id, name, sport, ... }]
POST	/api/athletes	Create new athlete	Add Athlete Modal	{ name, age, sport, position }	{ athlete_id, name, ... }
DELETE	/api/athletes/{id}	Delete athlete record	Squad Matrix	Headers: Bearer <token>	{ status: "deleted", athlete_id }
POST	/api/videos/upload-and-analyze	Analyze video file	Analysis Tab	FormData(athlete_id, activity, file)	AnalysisResult JSON
GET	/api/reports/{id}	Download PDF report	Analysis Screen	Headers: Bearer <token>	Binary PDF File Stream
GET	/api/health	Backend status check	Uptime Monitors	None	{ status: "healthy" }

SECTION 10: TRACE REAL DATA — "FOLLOW ONE VALUE" EXAMPLES

A. FOLLOW AN ATHLETE NAME ("Alex Rivera")

- * **User Input:** Typed in input box `<input name="name" value="Alex Rivera" />`.
- * **React State:** Saved in `form.name = "Alex Rivera"`.
- * **Save Click:** `handleSaveAthlete()` reads `form.name`.
- * **API Payload:** Sent as JSON `{"name": "Alex Rivera", "age": 22, "sport": "Football"}`.
- * **FastAPI Route:** `create_athlete()` in `backend/app/routes/athletes.py` receives Pydantic model `payload.name`.
- * **Database Write:** Instantiates SQLAlchemy `Athlete(name=payload.name)` and commits to SQLite.
- * **JSON Response:** Returns `{ "athlete_id": "ATH_8f9a2b", "name": "Alex Rivera", ... }`.
- * **React State:** `setAthletes(prev => [newAthlete, ...prev])`.
- * **UI Render:** Table row displays "Alex Rivera" in Squad Matrix.

B. FOLLOW A LANDMARK (`left_knee`)

- * **Detection:** MediaPipe BlazePose Index 25 detects `left_knee`.

- * **Coordinates:** Normalized 3D image coordinates $(x=0.482, y=0.615, z=-0.124, \text{visibility}=0.98)$.
- * **Vector Formation:** Linked with `left_hip` (index 23) and `left_ankle` (index 27) to form vectors $\vec{BA}$ and $\vec{BC}$.
- * **Calculation:** Vector dot-product formula calculates Left Knee interior angle as 92.4° .
- * **Series Rollup:** Aggregated across 30 frames $\rightarrow$ Min: 76.0° , Max: 156.0° , ROM: 80.0° .
- * **ML Input:** Feature `rom = 80.0` passed into Random Forest `.predict_proba()`.
- * **UI & Report:** Displayed on Kinematics Lab dashboard and rendered into ReportLab PDF telemetry table.

SECTION 11: VIDEO PROCESSING EXPLAINED FROM ZERO

- * **What is a video?:** A video is simply a sequence of still images (called frames) displayed rapidly one after another (e.g. 30 frames per second).
- * **Why process frame-by-frame?:** Computers cannot evaluate motion in a static file; OpenCV breaks the video into individual frame images to track joint position changes over time.
- * **How OpenCV reads video:** `cap = cv2.VideoCapture(file_path)` decodes compressed MP4 frames into RGB numpy array matrices.
- * **Stride Downsampling:** To prevent server lag, OpenCV resizes high-res images to 640px width and samples 30 representative frames across the video duration.
- * **What MediaPipe receives:** An RGB image array matrix of size 640×480 .
- * **What MediaPipe returns:** An array of 33 Landmark objects containing normalized (x, y, z) coordinates and visibility confidence scores.

SECTION 12: POSE ESTIMATION

Google MediaPipe BlazePose predicts key joint coordinates across the human body:

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(11) Left Shoulder (12) Right Shoulder



(13) Left Elbow (14) Right Elbow



(15) Left Wrist (16) Right Wrist



(23) Left Hip (24) Right Hip



(25) Left Knee (26) Right Knee



(27) Left Ankle (28) Right Ankle

* **Initialization:** `mp_pose.Pose(static_image_mode=False, model_complexity=0, min_detection_confidence=0.5)` in `backend/app/services/pose_estimation.py`.

* **Fallback Extraction:** If MediaPipe is unavailable in a minimalist cloud container, `_fallback_pose_extraction()` uses OpenCV Otsu binary thresholding and contour bounding geometry to approximate landmark positions safely without crashing.

SECTION 13: ALL MATHEMATICAL EQUATIONS

EQUATION 1: 3D Joint Angle Formula (`_angle_at_vertex`)

* **Formula:**

$$\vec{BA} = A - B, \quad \vec{BC} = C - B$$

$$\cos\theta = \frac{\vec{BA} \cdot \vec{BC}}{|\vec{BA}| |\vec{BC}|}$$

$$\theta = \arccos(\cos\theta) \times \frac{180}{\pi}$$

* **Code Location:** `backend/app/services/biomechanics.py` (Lines 72–83).

* **Explanation:** Measures joint flexion angle at vertex B formed by points A and C . Taking the dot product of normalized vectors yields the cosine of the interior angle.

* **Example:** If Hip $A=(0,4)$, Knee $B=(0,0)$, Ankle $C=(3,0)$, then $\cos\theta = 0 \rightarrow \theta = 90.0^\circ$.

EQUATION 2: Range of Motion (ROM)

* **Formula:**

$$\text{ROM} = \max(\theta_1, \dots, \theta_N) - \min(\theta_1, \dots, \theta_N)$$

* **Code Location:** `backend/app/services/biomechanics.py` (Lines 99–103).

* **Explanation:** Measures total angular movement distance from peak extension to peak flexion.

EQUATION 3: Bilateral Limb Symmetry Percentage

* **Formula:**

$$\text{Symmetry (\%)} = 100 \times \left(1 - \frac{|\text{ROM}_{\text{Left}} - \text{ROM}_{\text{Right}}|}{\max(\text{ROM}_{\text{Left}}, \text{ROM}_{\text{Right}})} \right)$$

* **Code Location:** `backend/app/services/biomechanics.py` (Lines 154–169).

* **Explanation:** Quantifies kinetic balance between left and right limbs. 100% indicates perfect symmetry.

EQUATION 4: Trunk Posture Lean Angle

* **Formula:**

$$\vec{S} = P_{\text{Shoulder_Mid}} - P_{\text{Hip_Mid}}$$

$$\alpha = \arccos\left(\frac{\vec{S} \cdot (0, -1)}{|\vec{S}|}\right) \times \frac{180}{\pi}$$

* **Code Location:** `backend/app/services/biomechanics.py` (Lines 172–219).

* **Explanation:** Measures spinal forward/lateral lean relative to vertical axis $(0, -1)$.

EQUATION 5: Machine Learning Scaled Risk Score

* **Formula:**

$$\text{Risk Score} = (P_{\text{ML}} \times 60.0) + ((100 - \text{Symmetry}) \times 0.20) + ((100 - \text{Quality}) \times 0.20)$$

* **Code Location:** `backend/app/services/risk_prediction.py` (Lines 200–230).

* **Explanation:** Combines Random Forest ML probability (60% weight) with biomechanical asymmetry (20%) and movement quality penalties (20%).

SECTION 14: "WHY DOES THIS EQUATION EXIST?"

* **Why not use raw pixel coordinates?:** Raw pixel values change depending on camera distance and phone resolution. Angles are scale-invariant and camera-distance invariant.

* **Why calculate symmetry?:** Studies prove bilateral limb asymmetry >15% directly correlates with ACL strain and compensatory muscle tears.

* **Why hard-code thresholds?:** Thresholds (e.g. 15% asymmetry limit, 25° trunk lean limit) are derived from published clinical sports medicine literature.

SECTION 15: CODE EXPLANATION

``python`

`backend/app/services/video_processing.py`

```
def validate_video_file(filename: str, file_size: int) -> None:
    ext = os.path.splitext(filename)[1].lower()
    if ext not in SUPPORTED_EXTENSIONS:
        raise VideoValidationError(f"Unsupported file format '{ext}'.")
    if file_size > MAX_FILE_SIZE_BYTES:
        raise VideoValidationError(f"File too large ({file_size / (1024*1024):.1f} MB).")
    `
```

* **What it does:** Validates uploaded video files before passing them to OpenCV.

* **How it works:** Checks extension against supported set (.mp4, .mov, .avi, .webm) and enforces 300MB size limits.

What to say: "This function acts as a security guard, preventing non-video uploads or oversized files from crashing the server."

SECTION 16: EXAMINER QUESTION → FILE TO OPEN

Examiner Question	File Path to Open in VS Code	Function Name	What to Point At
"Where is the ML model trained?"	<code>scripts/train_pipeline.py</code>	<code>train_injury_risk_model()</code>	Random Forest & XGBoost model training loop.
"Where are the saved ML files?"	<code>backend/app/models/ml/</code>	Binary Files	<code>injury_risk_model.joblib</code> artifact.

Examiner Question	File Path to Open in VS Code	Function Name	What to Point At
"How is ML risk predicted live?"	backend/app/services/risk_prediction.py	compute_risk()	.predict_proba() inference call.
"Where is MediaPipe pose tracking?"	backend/app/services/pose_estimation.py	PoseEstimator	mp.solutions.pose.Pose() initialization.
"Where is the 3D angle math?"	backend/app/services/biomechanics.py	_angle_at_vector()	Vector dot-product $\arccos$ formula.
"Where is video upload handled?"	backend/app/routes/videos.py	upload_and_analyze_video()	UploadFile processing and DB persistence.
"Where is password hashing?"	backend/app/routes/auth.py	login(), hash_password()	Passlib Bcrypt salt hashing.
"Where are the DB tables defined?"	backend/app/models/db_models.py	User, Athlete, Analysis	SQLAlchemy ORM class definitions.
"Where is the PDF generated?"	backend/app/services/report.py	generate_pdf_report()	ReportLab Flowable canvas document engine.
"Where is the frontend code?"	frontend/src/main.jsx	App(), useState()	React SPA components and theme context.

SECTION 17: EXAMINER QUESTION ENGINE (100 VIVA Q&AS)

Sample Viva Questions & Direct Answers:

Q1: What machine learning models were used?

* **Answer:** Random Forest Classifier and XGBoost Classifier ensemble models trained on 5,430 biomechanical samples, achieving 92.4% accuracy and 0.96 ROC-AUC. ([scripts/train_pipeline.py](#)).

Q2: Is password stored in plain text in the database?

* **Answer:** No. Passwords are encrypted using Passlib Bcrypt salt hashing ([pwd_context.hash\(\)](#)). ([backend/app/routes/auth.py](#)).

Q3: How is user authentication maintained across requests?

* **Answer:** Via signed 7-day JWT Bearer Access Tokens stored in [localStorage](#) and sent in the [Authorization](#) HTTP header. ([src/main.jsx](#) & [auth.py](#)).

Q4: How does OpenCV handle 4K video uploads without server lag?

* **Answer:** Downsamples frames wider than 640px to 640px and applies stride sampling, speeding up MediaPipe inference by 10x. ([backend/app/services/video_processing.py](#)).

Q5: What database is used in development vs production?

* **Answer:** SQLite locally ([backend/database/app.db](#)) and PostgreSQL in production, managed DB-agnostically via SQLAlchemy ORM. ([backend/app/database.py](#)).

SECTION 18: QUESTION ANSWERING METHOD

When asked an unexpected technical question by an examiner, use this 4-step emergency formula:

1. **Identify the System Component:** (Frontend, FastAPI Route, OpenCV/MediaPipe Engine, ML Predictor, or Database).
2. **State the Simple English Function:** Explain what that component does in one sentence.
3. **Open the Relevant File:** Use Section 16 to open the exact file and point to the line number.
4. **Explain the Code Input & Output:** Show the function inputs, transformation logic, and return value.

SECTION 19: "IF THEY ASK ME TO TRACE IT"

* **Trace Athlete Creation:** UI Form (`main.jsx`) $\rightarrow$ `POST /api/athletes` $\rightarrow$ FastAPI `create_athlete()` (`athletes.py`) $\rightarrow$ SQLAlchemy `Athlete` ORM model $\rightarrow$ SQLite `athletes` table $\rightarrow$ JSON response $\rightarrow$ React Squad Matrix re-render.

* **Trace Video Analysis:** Upload Button (`main.jsx`) $\rightarrow$ `POST /api/videos/upload-and-analyze` (`videos.py`) $\rightarrow$ OpenCV Frame Extraction (`video_processing.py`) $\rightarrow$ MediaPipe 33 Keypoints (`pose_estimation.py`) $\rightarrow$ Vector Angle Math (`biomechanics.py`) $\rightarrow$ ML Model `.predict_proba()` (`risk_prediction.py`) $\rightarrow$ SQLite DB Write $\rightarrow$ UI Dashboard Gauges.

SECTION 20: DATABASE EXPLANATION

* **ORM Framework:** SQLAlchemy 2.0.

* **Core Tables:**

- `users`: `id`, `name`, `email`, `role`, `password_hash`, `password_salt`, `created_at`.
- `athletes`: `id`, `user_id` (FK $\rightarrow$ `users.id`), `name`, `age`, `sport`, `position`, `injury_history`.
- `videos`: `id`, `athlete_id` (FK $\rightarrow$ `athletes.id`), `stored_filename`, `stored_path`, `processing_status`.
- `movement_analyses`: `id`, `athlete_id`, `video_id`, `biomechanics_json`, `movement_quality_json`.
- `risk_results`: `id`, `analysis_id` (FK $\rightarrow$ `movement_analyses.id`), `risk_score`, `risk_level`, `recommendations_json`.
- `sessions`: `token`, `user_id`, `created_at`.

SECTION 21: DEPLOYMENT DEEP DIVE

* **Frontend:** Deployed on **Vercel** as a static Single Page Application built via `npm run build (dist/)`. Configured with SPA rewrite rules in `frontend/vercel.json`.

* **Backend:** Deployed on **Render** running Python 3.14 and Uvicorn ASGI production server via `backend/Procfile: web: uvicorn app.main:app --host 0.0.0.0 --port $PORT`.

* **Environment Variables:**

- `VITE_API_URL`: Points frontend to `https://injury-prediction-backend.onrender.com`.
- `ALLOWED_ORIGINS`: Defines CORS allowed frontend domains.
- `DATABASE_URL`: Points SQLAlchemy to persistent SQLite or PostgreSQL storage.

SECTION 22: TIMING QUESTIONS & PERFORMANCE

- * **Website Initial Load:** 0.5–1.5 seconds (Vercel CDN static assets).
 - * **Cold Start Time:** 10–25 seconds if Render free-tier backend is sleeping.
 - * **Video Upload Time:** 0.5–2.0 seconds (depends on video file size and network speed).
 - * **OpenCV + MediaPipe Processing Time:** 1.2–3.0 seconds (optimized via 640px downsampling and 30-frame stride).
 - * **ML Inference Time:** 0.05 seconds (`joblib` pre-loaded in memory).
 - * **PDF Report Generation Time:** 0.3 seconds (ReportLab canvas generation).
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SECTION 23: ERROR / FAILURE FLOW

- * **Invalid File Format:** `validate_video_file()` raises `VideoValidationError(400)` if format is not `.mp4/.mov/.webm`.
 - * **Expired Token:** FastAPI dependency `get_current_user` raises `HTTPException(401)`. Frontend intercepts 401, clears `localStorage`, and redirects to Login.
 - * **Corrupted Video File:** OpenCV fails `cap.isOpened()`, raising `VideoProcessingError`. Backend logs notice and engages resilient kinematic fallback synthesizer.
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SECTION 24: PRESENTATION SCRIPTS BY DURATION

- 30-Second Script:** "Motion IQ is an AI-powered movement assessment system. Using standard smartphone videos, we apply OpenCV and MediaPipe to track 33 3D body keypoints. We compute joint angle vectors and pass these features into a Random Forest ML model to predict injury risk percentage and generate clinical PDF reports."*
- * **2-Minute Script:** Adds technical architecture details (React SPA on Vercel, FastAPI on Render, 3D vector geometry dot-product math, 92.4% ML accuracy, SQLAlchemy ORM database persistence, and ReportLab PDF engine).
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SECTION 25: LIVE DEMO GUIDE

1. Open `http://localhost:5173`. Show Login screen. Log in with test credentials.
 2. Navigate to Athletes Squad Matrix. Select athlete "Alex Rivera".
 3. Click "Upload Video", select sample squat video, click "Analyze Movement".
 4. Show real-time processing overlay, 3D skeleton pose tracking, and Risk Score gauge (78% HIGH RISK).
 5. Click "Download Clinical PDF Report", open generated PDF file showing biomechanical telemetry tables.
 6. Open VS Code to show `scripts/train_pipeline.py`, `biomechanics.py`, and `risk_prediction.py`.
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SECTION 26: CODE-TO-ACTION TABLE

Component	Event Handler	Frontend Function	API Endpoint	FastAPI Route	Backend Service
en>	handle Submit ()	api('/api/auth/login')	POST /api/auth/login	routes/auth.py	Passlib Bcrypt

Component	Event Handler	Frontend Function	API Endpoint	FastAPI Route	Backend Service
Modal	handleSaveAthlete()	api('/api/athletes')	POST /api/athletes	routes/athletes.py	SQLAlchemy ORM
Video Upload	handleUpload()	api('/api/videos/upload')	POST /api/videos/upload-and-analyze	routes/videos.py	OpenCV + MediaPipe
Report Card	handleDownload()	window.open(url)	GET /api/reports/{id}	routes/reports.py	ReportLab Engine

SECTION 27: FILE MAP

File Path	Primary Purpose	Key Functions	When It Runs	Show To Examiner?
frontend/src/main.jsx	Entire React SPA UI & State	App(), api(), AuthScreen	Always in browser	Yes (Frontend UI)
backend/app/main.py	FastAPI Server Entrypoint	FastAPI(), CORS setup	Server startup	Yes (Backend API)
backend/app/routes/auth.py	Authentication Endpoints	login(), register()	User Login / Signup	Yes (Auth & Security)
backend/app/routes/videos.py	Video Upload Router	upload_and_analyze_video()	Video submission	Yes (Video Pipeline)
backend/app/services/video_processing.py	OpenCV Frame Extraction	process_video()	Video processing	Yes (Computer Vision)
backend/app/services/pose_estimation.py	MediaPipe 33 Keypoint Pose	PoseEstimator	Frame processing	Yes (Pose Tracking)
backend/app/services/biomechanics.py	3D Vector Geometry Math	_angle_at_vertex(), compute_symmetry()	Kinematic analysis	Yes (Math Formulas)
backend/app/services/risk_prediction.py	ML Inference Engine	compute_risk().predict_proba()	Risk calculation	Yes (ML Prediction)
scripts/train_pipeline.py	ML Model Training Script	train_injury_risk_model()	Offline model training	Yes (Model Training)
backend/app/services/report.py	PDF Document Builder	generate_pdf_report()	Report download	Yes (PDF Generation)

SECTION 28: WHY DID WE USE THIS?

- * **React vs Plain HTML:** React provides Virtual DOM re-rendering, component modularity, and smooth tab switching without page reloads.
- * **FastAPI vs Django:** FastAPI is lightweight, asynchronous, faster, and built specifically for REST APIs.
- * **MediaPipe vs OpenPose:** MediaPipe runs in real time on standard CPUs without requiring heavy NVIDIA GPUs.
- * **Random Forest vs Deep Neural Nets:** Random Forest prevents overfitting on tabular biomechanical datasets and provides transparent feature importance rankings.

SECTION 29: SYSTEM LIMITATIONS

- * **Optical View Requirement:** Requires unimpeded visual line of sight to athlete limbs.
- * **Lighting Dependencies:** Extreme shadows or low lighting decrease MediaPipe keypoint confidence.
- * **Screening Purpose:** Motion IQ is a movement screening tool. **It does NOT provide medical diagnosis or replace licensed physicians.**

SECTION 30: COMPLETE END-TO-END STORY

"The user opens our React Single Page Application on Vercel. The browser downloads JavaScript assets and mounts the UI. When the user logs in, React sends an HTTP POST request to our FastAPI backend on Render. FastAPI verifies the password hash via Bcrypt and returns a 7-day JWT Bearer token. Next, the user selects an athlete and uploads a 5-second squat video. FastAPI saves the video file to disk and passes it to OpenCV. OpenCV downsamples the frames to 640px for rapid execution. Google MediaPipe BlazePose detects 33 3D skeletal keypoints per frame. Our vector geometry engine calculates 3D joint angles using the vector dot-product cosine inverse formula. It computes range of motion, trunk lean, and knee symmetry. These 23 kinematic features are passed into our Random Forest ML model, which outputs a 78% Risk Score (HIGH RISK). The backend stores the analysis in SQLite via SQLAlchemy ORM. Finally, FastAPI returns JSON to React, which updates the UI dashboard with visual risk gauges and allows the user to download a complete clinical PDF report built by ReportLab."

SECTION 31: EMERGENCY "EXAMINER CAN POINT AT ANYTHING" GUIDE

- * **If examiner points at a Button:** Open `frontend/src/main.jsx`, show the `onClick` event handler function.
- * **If examiner points at an API URL:** Show `API_BASE_URL` in `frontend/src/main.jsx` and the corresponding `@router` decorator in `backend/app/routes/`.
- * **If examiner points at a Formula:** Open `backend/app/services/biomechanics.py`, show `_angle_at_vertex()` vector dot product.
- * **If examiner points at a Risk Score:** Open `backend/app/services/risk_prediction.py`, show `_KINEMATIC_MODEL.predict_proba()`.

SECTION 32: ZERO-KNOWLEDGE DICTIONARY

- * **React:** A JavaScript library for building component-based user interfaces.
- * **FastAPI:** A high-performance Python framework for building REST APIs.
- * **MediaPipe:** Google's AI framework for tracking 33 3D body pose keypoints.

- * **OpenCV**: A computer vision library for opening and processing video frames.
- * **Random Forest**: An ensemble machine learning algorithm based on decision trees.
- * **SQLAlchemy**: A Python ORM that translates Python code into database SQL queries.
- * **JWT**: JSON Web Token used for secure user session authentication.
- * **CORS**: Security rules controlling cross-origin API requests between web domains.

SECTION 33: QUICK REVISION CHEAT SHEET

- * **System**: Sports Injury Risk Detection (Motion IQ)
- * **Frontend**: React 18, Vite, Vercel Host
- * **Backend**: FastAPI (Python 3.14), Render Host, Uvicorn ASGI Server
- * **Computer Vision**: OpenCV + Google MediaPipe BlazePose (33 Keypoints)
- * **Vector Math**: $\theta = \arccos\left(\frac{\vec{u} \cdot \vec{v}}{|\vec{u}| |\vec{v}|}\right) \times \frac{180}{\pi}$
- * **Symmetry Math**: $100 \times (1 - \frac{|\text{Left} - \text{Right}|}{\max(|\text{Left}|, |\text{Right}|)})$
- * **ML Models**: Random Forest + XGBoost (92.4% Accuracy, 0.96 ROC-AUC)
- * **Database**: SQLite / PostgreSQL via SQLAlchemy ORM
- * **PDF Generator**: Python ReportLab Engine

SECTION 34: MEMORIZE THIS — THE 1-PAGE SUMMARY

"Our project is an AI sports injury risk detection system. The React frontend hosted on Vercel sends video files over HTTPS to our FastAPI backend on Render. OpenCV decodes the video frames, and Google MediaPipe BlazePose tracks 33 3D body landmarks. We use 3D vector dot-product cosine inverse math to calculate joint angles, bilateral symmetry, and trunk lean. We feed these 23 movement features into a Random Forest machine learning model to calculate an exact injury risk percentage. The results are stored in a database via SQLAlchemy ORM, and ReportLab generates downloadable clinical PDF reports. That is how our system works from end to end!"

SECTION 35: FINAL ACCURACY AUDIT & FACT SHEET

* Verified Actual Code Features:

- React SPA with 8 UI Theme Switcher (`frontend/src/main.jsx`).
- FastAPI backend with 7 route modules (`backend/app/routes/`).
- OpenCV frame decoding and 640px downsampling (`backend/app/services/video_processing.py`).
- Google MediaPipe BlazePose 33 keypoint extraction (`backend/app/services/pose_estimation.py`).
- 3D vector geometry dot-product angle math (`backend/app/services/biomechanics.py`).
- Random Forest + XGBoost ML inference (`backend/app/services/risk_prediction.py`).
- ReportLab PDF generator (`backend/app/services/report.py`).
- SQLite / PostgreSQL persistent database storage with SQLAlchemy ORM (`backend/app/models/db_models.py`).

* Features Not Implemented in Current Project:

- Live real-time audio voice coach correction during exercise execution (browser TTS text-to-speech briefing is available, but live exercise voice correction is NOT IMPLEMENTED IN THE CURRENT PROJECT).
- WebSocket live video streaming (video analysis operates via HTTP POST file uploads).
- Docker backend container on Vercel (Vercel hosts frontend SPA only; backend runs on Render).

*** Deployment Facts Verified:**

- Vercel rewrites all non-API routes to `index.html` (`frontend/vercel.json`).
- Render executes `uvicorn app.main:app` (`backend/Procfile`).
- Production backend API URL: <https://injury-prediction-backend.onrender.com>.